Pavanitha Manche

MS Computer Science @ USC

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Education

University of Southern California — Los Angeles, California
Masters in Computer Science (**GPA - 3.76/4.0**) — 2026 – Present

Indian Institute of Technology (IIT), Dharwad — Dharwad, India
Bachelors of Technology in Computer Science and Engineering (**GPA - 8.74/10.0**) — 2020 – 2024

Experience

Student Developer (Full-Stack & Data), *Keck Medicine of USC, Los Angeles* — May 2026 – Present

- Built full-stack C#, .NET and SQL Server features for Beacon, a finance platform for 5,000+ employees.
- Developed an ETL pipeline that loads SAP Concur data into an SSAS Tabular model with DAX, powering Power BI dashboards where department heads track employees' daily travel-card charges.
- Integrated MCP across multiple tools for query generation, schema navigation, and financial analysis.

Software Engineer (Backend & IoT), *ABB Innovation Centre, Bangalore, India* — May 2024 – Dec 2025

- **Sensi+:** Architected an industrial analyzer platform using C#, .NET and Angular. Designed analytics, caching, and REST API layers over PostgreSQL and MongoDB, deployed on Kubernetes.
- Built scheduler-driven MQTT services to ingest telemetry from over 10,000 analyzers, precomputing real-time aggregates and caching them in Azure Blob Storage to reduce data-load times by ∼5 seconds.
- **Gas Lift Optimization:** Developed a .NET backend for real-time data ingestion and ML-based setpoint generation for oil wells, with cron-scheduled jobs processing wells on parallel threads.

AI Intern (GenAI / RAG), *Dvara Solutions, India* — Jul 2023 – May 2024

- Built a voice-enabled GenAI RAG assistant in 6 Indic languages for low-literacy rural finance users.
- Designed the retrieval pipeline, semantically chunking unstructured data into a vector database and using similarity search to feed the most relevant chunks into the LLM prompt to ground its answers.
- Added session memory and prompt engineering for coherent chats, with TTS replies in local dialects.

Selected Projects

Rewind: Durable Execution for LLM Agents | *LangGraph, PostgreSQL*

- Architected a durable-execution layer for agentic LLM workflows in LangGraph, enforcing exactly-once tool calls via a PostgreSQL write-ahead ledger, idempotency keys and fencing tokens.
- Integrated Stripe and SMTP tools for AI agents with outcome reconciliation and precondition re-validation, validated via fault-injection testing against LangGraph, Temporal and DBOS.

Voice OTP based Authentication System | Papers: SPECOM 2023, IEEE SPCOM 2024

- Developed a two-factor Voice OTP system combining ECAPA-TDNN speaker verification and Conformer-based digit recognition, achieving 93.21% accuracy, resulting in 2 publications.
- Optimized on-device inference for mobile by benchmarking latency and memory across devices.

Achievements & Leadership

- Received ABB X-tra Miler Award twice (Q4 2024, Q3 2025) and the ABB Team Excellence Award for outstanding performance and successful project delivery.
- Mentored 3 Master's students on academics and campus life via the USC Viterbi Mentorship Program.

Skills

Languages: Python, C++, C#, Java, JavaScript, SQL, Bash / Shell Scripting
Generative AI: LLMs, RAG, AI Agents, MCP, Prompt Engineering, LangChain, Vector Databases
Machine Learning: PyTorch, TensorFlow, NLP, Speech Recognition, Model Optimization
Full Stack: .NET, React, Node.js, REST APIs, SQL Server, PostgreSQL, MongoDB
Data & DevOps: Power BI, SSAS, GCP, Azure, Docker, Kubernetes, Jenkins, Git, Linux / Unix

Relevant Courses

Computer Science: Natural Language Processing, Analysis of Algorithms, Web Technologies, DataBase Systems, Automata Theory, Computer Architecture, and Software Development.